

ActInf GuestStream #011.1 ~ Gregg Henriques

GuestStream | Gregg Henriques | 1:43:38

hello everybody welcome to the active inference lab today it's october 22nd 2021 and it's actin flap guest stream number 11.1 i'm daniel i'm here with blue and our guest greg so greg thank you so much for being here and please introduce yourself and take it away we'll be monitoring the live chat so if you have any questions during the presentation please feel free to ask them and we'll talk about them at the end but otherwise thanks again for being here and looking forward to this event wonderful hey it's real honor to be here uh i've seen a couple of other presentations they're really fascinating i look forward to sharing some of my version of reality um so let me give you a little bit of orienting coordinates for me um so i am professor uh in the combined integrated clinical and school doctoral program at james madison university uh where i've been since 2003 uh the role i function there is to train individuals to be psychological doctors is the term we use uh the reason we use that is because in psychology in the profession side there are different practice areas uh clinical counseling and school are the dominant

practice areas but we want to transcend
those into a more
general combined integrated
perspective
that affords us a way to kind of get out
of some of the old
traditional institutional silos and
get a more general integrated picture so
i've been there since 2003
my journey in this is

i
went to undergrad got my major in
psychology and was just going to be a
behavioral scientist
that wanted to be a clinician
my sense was the right thing to do was
to apply the methods and findings of
science to the clinical world
i went off got my master's in clinical
and community psychology
in 1994.

and um
well why don't i go ahead and start the
presentation and then i'll walk you
through some of that history and then
we'll get to this scenario where
ultimately you get this weird cartoon of
a garden on behind me and
we'll get to there over a period and
then we'll start into
what we're actually dealing with let me
share my screen here

so
so we're talking about the ontology of
mental behavior um
and uh
just to kind of give it a context um
my basic point for defining this talk

is then to be in contrast to
the current mainstream approach in
academic psychology
the mainstream approach in academic
psychology is really best frame is what
i would call
methodological behaviorism
methodological behaviorism term by jay
moore
but all it means is basically what is
psychology in the academy right now is
that it applies the methods of
behavioral science
to a thing
very vaguely defined as behavior and
mental process
okay
and indeed the methods of science really
are what end up defining the term
behavior which is the you can see
from a science perspective okay
uh and then the mental processes are
sort of the stuff that's inferred that
give rise to
the change in behavior at least if
you're a mentalist or a cognitivist in
traditional psychology as opposed to say
a radical behaviorist okay so the
science of
psychology is really defined
by an institution that applies methods
of science methods of behavioral science
in particular
to analyze behavior and then develop
models of mental process that afford the
explanation
now the structure of it is broken at the
level of coherence and what i mean by

that is it doesn't there's no organizing
framework for psychology

uh psychology is not defined by and
never has been defined by a paradigm
meaning a full in a cuneian sense
paradigm meaning a full like newton
developed a paradigm for classical
mechanics that then established
psychology has had is a is a field that
had lots of different schools of thought
many paradigms that tried to manifest
itself as sort of the dominant paradigm
that didn't happen

um there were no dominant paradigms in
psychology uh it gave rise to what's
called the crisis in psychology that led
vygotsky specifies in 1927. so we're
talking now

over

uh almost 100 years ago in fact it was a
first identified in 1899 the crisis

um and psychology then just punts on the
issue it's like what do you what do you
mean by the what's the field subject
matter and the answer is well

different schools of thought we'll frame
that in different ways okay

the ontology of mental behavior is the
argument that actually we can do better
than that

um we don't have to define the field
based on epistemology i.e the methods of
science but we actually can
define the field based on a scientific
ontology that clarifies what is meant by
mental behavior

and in so doing it affords a coherent
solution to the problem of psychology so

the problem of psychology is the inability of the field to cohere around a particular conception of its subject matter so that's the basic structure um i'm like i said i'm at james madison combined integrated so what's the basic gist just to get a lay of the landscape from where i'm coming from and i'm you know i'm an odd bird so we should be very clear that i i've been doing a weird version of psychology now for uh for a long time i started to say this i entered into the field sort of in a conventional sense i got trained in the normal methods of behavioral science and analysis of variants and and all that good stuff and it's like oh well clearly we want to employ a standard empirical method to delineate you know what the sort of the truth is or the most accurate models um and so uh but then i encountered this thing called the problem of psychotherapy uh what's that we actually get into the room with people and that's that's a trick um and then you're a real person in a real situation in the idiosyncratic processes by which you have to make active inferences about what is operating in the moment okay um means that the generalized aggregate variable analysis that the rest of the field does like oh here's a problem it's called major depressive disorder here's a

specified intervention called cognitive therapy and now you're to imply this cognitive therapy intervention with this person um sorry but what i learned and i didn't know it at the time but what i learned was the level of abstraction at the variable aggregate level relative to the actual process by which you engage in feedback in the room in a participatory real relationship of course i have language for this now that i didn't have them but the bottom line was god you know uh what i actually saw in the field was that the best of the best depending regardless of the school of thought the best of the past did the best surprisingly enough and lousy players from various all the different schools did lousy and most people did average and the schools of thought really don't generate um systematic difference at least the bona fide schools of thought so essentially what you get in psychotherapy is what's called the dodo bird finding uh i'm president-elect of the society for the exploration of psychotherapy integration which is a field that has long standing questions in relationship to the relationship between the schools of thought um and uh and and i got then introduced in 1994 this is basically to the field through a potentially integrated perspective so for me that was fascinating um it turns out that my orientation is actually

what's called a big picture coherentist orientation meaning that i actually want to zoom way back

organize pieces in a particular way so that i can create a coherent frame of reference and then zoom back in with the coherent frame of reference that's the way my mind operates so i had zoomed out and then realized there was actually no coherence um at the level of psychotherapy and people had known this for a long time and then this whole field that emerged in the mid 1980s this was like you know 15 years 10 years down the road from that emergent of that field with the psychotherapy integration problem

and then i was fascinated by it and i did something that's actually no one's ever done which is kind of interesting um i looked at medicine and this is just all intuitive i looked at medicine um and i said you know

you don't have all these different schools of thought in medicine it's not like you have you know the the cardiologists are not battling with the endocrinologist um versus uh you know the people that emphasize uh the gut you know they all are like hey they're different systems of you know biological adaptation that function together in a hole and think we have a thing called human biology

human biology is the science that organizes the various domains um and that affords a descriptive explanatory framework that then should

inform the practice of medicine so if we jump up a level it seems reasonable to me or it seemed intuitively reasonable to me that there should be a human science of human psychology that should organize the descriptive explanatory network of understanding okay and then apply that to the field all right so now notice what i had done i got socialized into the empirical methodological behavioral science approach of psychology dropped into the problem of psychotherapy backed up and was like oh wait a minute what i really need as a practitioner is i need a holistic conception of human psychology that if ford's clarity about what the hell the field is actually talking about sorry i'll probably drop an f-bomb here or there if that's okay but anyway what i basically realized then here is i shifted perspective and i said well what is psychology and then i realized oh my god nobody knows what the psychology is there's no word that everyone's got different reference points for this fundamental word okay and that's the problem of psychology there's a dramatic set of reference points um that and so i then uh took a couple of left turns that i can tell you about in 1996 and then a big one in 1997 and all of a sudden i actually developed a vision um not like some transcendent vision although it has these elements i developed a picture an organizing picture really a new map of big history as it were

that afforded me a new way to address
this problem of what psychology is okay
and that ultimately gives rise to a
clear ontology of the mental all right
we'll specify what i mean by that and
then by affording clarity about what the
scientific conception of the mental is i
could then uh organize my work and
return to the original problem and
develop a framework for psychotherapy uh
and so that's what the project is about
now my journey over this time period
basically is the first thing is hey
here's a problem and then i need this
and then i built this thing called the
tree of knowledge
system the tree of knowledge system is
this natural scientific ontology that
maps the world in a particular way that
affords a science
of a natural science view of what the
mental is
um i worked on that and said hey it
organizes the field um i did a you know
i can tell about my publication history
etc
and then ultimately that evolved in
what's called the unified theory of
psychology
the unified theory of psychology
specifies the tree of knowledge and then
a couple of other key ideas
that then provide a conceptual
architecture for assimilating and
integrating the schools of thought and
major empirical findings into a coherent
whole
i'll say again it provides a conceptual

architecture for assimilating and
integrating the schools of thought and
the major major schools of thought and
major empirical findings to afford a
coherent poll of the picture
and basically then
create a new unified theory of
psychology
i then shifted that i did that that
published that in 2011.
um i then returned to the original
problem um and from 2011 2016 or so
worked out then given this framework
what would a unified approach to
psychotherapy be
and i then delineated four key ideas
that could help organize the field and
generate a unified approach to
psychotherapy at least with individual
adults okay childs are different blah
blah blah i was honed in on sort of
standard individual psychotherapy
especially for neurotic adults which are
the vast majority of people come in for
psychotherapy or neurotic adults okay so
this forwards then a one-on-one a way to
organize the key insights and then and
so then you would have a unified theory
of psychology and a unified approach
okay
well then in 2016 as i was delineating
that i took another weird turn all right
um and there are a lot of reasons as to
why uh but um the short story is as i
was at a conference uh called
cultivating uh the globally sustainable
self
um about transformative education and

the massive global problems that we face
in relation um and it was in the context
of that conference

um i had created this model the unified
theory and unified approach it was
called a tua okay that's it just dropped
into the language you know how things
would get more and more efficient you
know it's ut and ua and you tattoo uh
somebody's like what the hell's your two
lines like how's this video blah blah
blah

so i'm at this conference and they're um
uh doing this talking fundamentally
about how to do this thing called
cultivate a globally sustainable self
meaning how do we transform our
education for children in the
world that we're in given all the crises
and challenges that we face what does
that actually look like it was
fascinating conference

and at the end of it there was a panel
discussion okay and panel discussion
actually got all the people they invited
there and they came up and they actually
all fumbled over the question as far as
i was concerned like they were brilliant
people they did this they couldn't
actually there was there's a summary
opportunity i left the conference and i
was walking away and i was talking to
somebody i was like you know what we
should do we should plant it to a seeds
and grow it to a trees
okay

um which sort of fell out of me as a
light bulb moment and then turned into

the thing behind me which is this garden
so all of a sudden i basically then
transformed this build this construction
that i had with all these different sort
of like a cathedral that got transformed
into a garden i actually then got called
into like spiritually artistically to
build this garden
so that now this knowledge system then
could be represented
symbolically in a garden and that's what
spine is a it's a it's a layered
knowledge system so like for example
well i'll just so that that's that's and
then you know i got pulled more into
that and then that shifted me further
and further into the orbit of weirdness
all right but anyway that's basically
what i realized fundamentally is why i
got pulled into this level of
abstraction um was something that i sort
of
got clear on as i've been writing my
current book
and that is a real understanding
fundamentally of what's going on here in
relationship to kind of our knowledge
system so we have a foundational problem
in our knowledge and what i call in this
foundational problem is called the
enlightenment gap what the hell is that
the enlightenment gap is the point that
when science emerged okay um in from
galileo and newton coming off of the
scholastics okay when we built physical
science and the epistemology and
ontology that emerged with physical
science um there was never a ability to

take what we learned from the physical sciences and create a holistic synthetic philosophy can't tried hegel tried etc but what you basically get in the enlightenment is a gap a failure to get the proper relationships first on the relationship between matter and mind okay some base mind body problem what the hell is that how do you actually solve that and then the other thing that emerges as a secondary joint problem is what is actual scientific knowledge relative to social knowledge okay and this problem emerges we see in the 20th century with the emergence of the philosophical post-modern critique of modernity which basically says hey modernity made too many abstract pure reason progress claims that they were actually that you could find genuine truth that's not contextualized in a socio-historical justification context that's the post-modern critique and actually you can't really make knowledge claims that are transcendent they're all imminently contextualized in the socio-historical context and lo and behold look at the way the knowledge systems emerge they get they privilege cis-gendered white colonial men who go around and create oh yes this is true reason and behold it's actually it's normative for us and we get to distribute our power structures around the world and so you get a philosophical and a social justice

post-modern critique about what science
is relative to social knowledge that
never actually gets resolved okay that's
why we have post-modern versus modern
onto epistemological debates about the
nature of knowledge and so the
enlightenment gap basically is a
description of the current state of
affairs where we have we're lacking a
philosophical framework that affords us
clarity about matter and mind on the one
hand and science and social knowledge on
the other
okay

so that's the gist of the location about
where we are
um

i built this thing called the unified
theory of knowledge okay
what is unified unified is taken
directly from the book consilience the
unity of knowledge e.o wilson wrote a
book called consilience in 1998 called
the consultancy unity of knowledge
that's what unified means it means
conciliating there's a consilient
coherent big picture view that actually
gets the pieces of the puzzle correct so
we can actually understand
knowing

okay and knowing how we know and knowing
what we mean when we know things
physically and what we know things
biologically what we actually mean
psychologically human psychologically
into the social sciences how we know
scientifically relative to how do we
know subjectively

what do we know about ethics etc there's actually a coherent synthetic philosophy that is available that frames these problems that wasn't achievable before um

and so here's the garden okay uh and what it has here is this so for example right here this little thing's called the stepping stone what is that s-t-e-p-p stands for the standard theory of elementary particle physics and the stepping stone then grounds you in physics in a particular way then this first branch is the tree of knowledge system the base of which is the energy information quantum field structures that live underneath matter and go back to big bang

and then each one of these is a layered system of information processing called life mind and culture

and it is that arrangement of energy information gives rise to a classic material field and then life emerges and then mind and i'll explain what i mean by mind here emerges at the animal mental behavioral level and then culture emerges so that we become go from primates to people as we talk to each other and build what are called systems of justification

okay so that just gives you a flavor for it there's other things that are going on here that i won't necessarily get into but it basically says there's actually a knowledge system here the left side is the unified theory you can see actually there's a little u here

and a t and these four ideas make up
what's called the unified theory there's
a u and an a which stands for the
unified approach
okay this right here is a symbol of
mindfulness
on the right side of the branch is the
way in which we relate to people in
healthy psychological ways it's actually
called calm mo it's an integrative
approach to psychological mindfulness
it's the processes by which we can hold
each other in ourselves and deal with
difficult socio-emotional relations that
afford productive virtuous looping as
opposed to what often happens people as
they get trapped in negative cycles
attack each other and then get in
vicious destructive looping that's just
one example so the tree then has eight
key ideas four of which make up the
unified approach this is by the way the
tree of life the tree of knowledge is
the first branch tree of life has four
here four here and then as this symbol
in the middle which is actually called
the metaphysical empirical flower
metaphysical empirical flower creates a
dialectic between metaphysics and
empiricism and that was missing in the
structure of the enlightenment which we
need there's another thing down here
called the coin but i'll pause here just
so there's plenty of um structure there
let me make a few more comments and then
actually i'll pause after this so
ultimately what the unified theory does
is it affords a new opportunity to think

about the enlightenment gap and i are
you resolve it um and what it does is it
sets the stage for a unified theory of
psychology
and frames a unified approach to
psychotherapy okay
um so that's basically then we can get
into what does all this mean in
relationship to some of the deep theory
but i wanted to give folks an
opportunity for the basic gist of where
i'm coming from
thank you for the summary cool
work and symbolic images
so anyone watching live can definitely
write questions but that's a great place
to stop we got the personal
and the multi-level narrative
coming together with a little bit of a
challenge and a response
which we'll then learn more about so
maybe we can just go blue if you have
any comments otherwise we can go into
the next section
um i don't know if we want to go through
some of my questions now i mean i have
some that relate to the uni unifor
universal theory of knowledge um so if
we want to maybe pull up that slide or
if uh greg is still sharing slides we
don't have to
i cannot just read it off this slide so
the first perspective yeah and it's on
the screen as well so here's um uh one
just summary quote
first the the ut okay differentiates the
mental behavior of animals from the
behavior of other living creatures both

metaphysically and metatheoretically
which you mentioned and second
it differentiates the mental behavior of
human persons from the mental behavior
of other animals both metaphysically and
metatheoretically okay so
similar but a little bit different there
so what are the distinctions that are
critical for psychology
what distinctions does your work make
how do we know those are the right ones
how did you carve
it out okay
at the jigsaw joints
beautiful okay so yeah i love that
that's a wonderful set of questions and
right gets right to the heart of some of
the key issues okay so let's talk about
the so the the tree of knowledge divides
is a descriptive metaphysical system
that's a um
the picture is and it shows matter life
mind and culture okay so basically
coming off big bang so you start at
what's called an energy information
singularity all right that's how you go
back to big bang 13.8 billion years ago
et cetera you for you run the tape back
and everything collapses into
quote-unquote a point or a singularity
because this is just general cosmology
all right i'm getting to the question
but i just give this background okay so
according to modern cosmology the
universe essentially begins in a state
of pure energy information that's at
least what i refer to it as we can talk
about the term and then out of that

comes matter

okay uh and basically what you get is
you get all the four fundamental forces
in nature that's electromagnetism
gravitation strong weak
differentiating to create fields and
then you get a coalescence a cooling and
freezing and coalescence of energy into
chunks of matter in the form of
particles atoms up into them chemicals
in certain unique contexts and then
across scale that builds things like
stars and galaxies and mega galaxy
clusters okay

so this is then our physical universe
all right and notice what i've done and
the reason i had to say this is because
one of the first things we have to note
from classical mechanics is that modern
physicist physics teaches us that
there's a thing beneath
the classical material world called
quantum field theory and the energy
information singularity of big bang
the reason that's really important
because we have to understand the
fundamental substance ontology of the
world which is energy information fields
not billiard balls

okay so they were working in the initial
sort of ontology is billiard
and it's like no that's actually not the
ground of the universe ground of the
universe is sort of like an energy
information super force field theory all
right which by the way for mental
implica the implications to understand
the way mental structures work it's it's

important that some people like bernardo
castro drop it all the way around call
that big mind i don't agree with that
but fundamentally we do want to
understand that okay
then we come and we jump up
we don't know how this exactly happens
but about 4 or 3.8 billion years ago we
see the jump in
the world from chemicals complicated
chemical protein lipid structures that
someone get jumped together organized
and we get life
okay so the miracle of life
all right
so the miracle of life
then creates the jump from
quasi-organizing systems that then get
into fundamentally slooping into
self-organized
what
deacon
terence deacon calls an intentional
phenomena that operates across abtentionia
okay so intentional phenomena meaning
that basically there is some
descriptively not explanatory but
descriptively these cells behave with a
goddamn purpose
they're doing okay
and it somehow is that they're imagining
that they are black and then they move
toward the lack like they're hungry and
they
and they don't want to die and they move
around okay incomplete nature is terence
deacon's analysis of this jump
well this is really important to

understand the emergence of complex adaptive dynamic systems because that's what life is it's a complex adaptive dynamic system that has embedded in it energy information systems that afford the capacity to pull

i'll use a slightly different color first and free energy from the uh environment to do work to maintain its complex organization so the survivor reproduces

okay

and the reason i have to tell you all of that is because the ground at what we need to understand the bi dynamic biological ground out of which then minded animals are going to emerge

okay

so so that we get a natural selection operating on genetic combinations and what happens according to unified theory is actually you get an orthogonal dimension of activity orthogonal to the real in some ways what do i mean by that it's the subjective information

processing the epistemic information processing of a cell

okay it's like the cell is actually making judgments and quasi even decisions as it engages in some kind of primitive act of inference

okay

um and so then that becomes the ground and then basically you get bacteria that are doing that for a long time and other kinds of animal kingdom i mean uh you know algae and other kinds of protists kind of kingdom stuff

and then you get multicellular plants
okay um that afford the opportunity for
what big historians call big life so you
get
cellular plants and you're also going to
get fungi
um mycologists study things like they do
really weird stuff uh fungi um and then
you get so you're gonna get these active
multicellular entities
that are still though fundamentally
organized and grounded
and change their environment through
their growth potentials
within the cell so even things like
mushrooms what mushrooms don't do is
they don't get up and walk around
all right um but what they do is they
send out growth signals into particular
things and then they grow in really
interesting ways but they are confined
by their boundaries
all right
but what happens in the
in the animal kingdom
as a shoot-off in relationship to this
so that this system now becomes as a
multi-organizing system becomes mobile
all right so 700 million years ago
you're going to get then a transition
into
basically jellyfish-like creatures okay
that have distributed neural networks
that have some degree of capacity for
coordinating their behavior
all right through the distributed neural
net
that's the first step towards a minded

animal it's not there yet okay

you then get bilateral planning with an
in an emerging integration between the
sensory motor structures okay like in a
planarial worm

now a planaria worm's got a centralized
controlled center okay that's then
distributing and it's got bilateral
movement to coordinate its movement in a
particular way now you're very close
with a planaria worm to what i would
call a minded animal

the final thing is that you then got the
cambrian explosion

550 520 million years ago and in this
short period of time basically you get a
much greater complexification of the
nervous system you get a centralized
nervous system and you get a complex
active body plan

that complex active body plan then
regulates a sensory motor looping
okay that affords the animal to get up
and move around

all right now that's a game changer it
moves the entire structure as a sensory
motor loop

okay and it is that division that joint
point

okay between life living organisms and
minded animals that the tree of
knowledge points to and says hey that
cambrian explosion thing is when minded
animals blew out of living

organisms there's a joint point thing
that's as important to understand in our
big picture as the joint point between
non-inanimate matter and matter

and it follows a very similar pattern
because what you see like with cells
what did you get you get an rna dna
information processing closure around a
lipid with protein generation that then
creates a communication network between
other cells

well the nervous system is very much
akin to the rna dna it's a centralized
control system and what it does is it
yokes all the cells together in a
coordinated way so the animal gets to
move as a whole unit

so the sensory motor looping of animal
movement then becomes a fundamental
another dimension of information
processing communication and we get
animal animal movement at working with
each other in a communication network
you get the base of the mind dimension
okay so in the unified theory that's and
you can and then it's defined
behaviorally and descriptively so it's
just a complex active bodies moving in
relationship to each other with
functional

outcomes
that's what it is so when you go and you
walk your dog your dog is exhibiting
this the insects are exhibiting this
okay

and it's it's called mindedness or mind
it's the functional animal the behavior
as a whole mediated by the nervous
system okay it produces a reliable
understandable effect the base of this
looping i see as basic you know
predictive processing active inference

essentially okay there's a fundamental iterative process that is regulating information energy in particular structures that theory the meta theoretical structure i bring to bear is called behavioral investment theory okay behavioral investment theory is a meta perspective that takes together what's called energy economics so you have to solve the problem of uh entropy and energy expenditure so first principles energy economics second is evolution the natural by natural selection and other any other things we have to add to that but that's the general gist the third principle is behavioral genetics which means the genetic the gene system is going to bias the system in a particular way the fourth principle is neuro-computational control so neural computational control then the processes by which we engage in neural computational control then is analyzed deeply by an active inference predictive processing the fifth principle is learning and of course then that's how a system upgrades in relationship to experience and the sixth principle is life history development meaning depending on where the animal is in its life history is it young is it adolescent is it late life et cetera it's reproductive and survival strategies are going to change based on its life history development so those are the six principles of behavioral investment

and what they do then is they yoke
together neuroscience and evolutionary
theory and computational cognitive and
by the way behavioral science from the
vantage point of a radical behavioral
position like a skinner
this is actually i come through this
skinner by the way which is weird okay
so so skinner actually what skinner does
is he just sits on the outside he's he
basically said oh it's black box
it's physiology black box but what i
will track is i will track at a very
fine-grained analysis consequences the
animals doing something has got
consequences and it is the consequences
that pull the animal in its behavioral
investment and i can navigate the
consequences
as an instrument as an engineer he
thinks like an engineer he's not
thinking like a theorist he's really an
engineer you know you want to be a poet
okay but it basically fails as a poet
and he's like well what really but you
know really we can do a psychodynamic
interpretation what happens to him but
actually then he starts just focusing on
how to control the consequences to pull
investment from an animal and so that's
an exterior behavioral position and
that's radical behaviorism which by the
way is totally different than
mythological behavior and it's a very
you know five percent psychologists go
this week applied behavior analysis but
it's really key because we want to be
able to bring the fact of the matter is

the dynamic control consequential system
from the outs behavioral outside is
going to connect to the after active
inference so somebody asks there's
questions like hey can we connect
enactment and an activism and
representationalism with a behavioral
selection structure yeah yeah you
can okay so anyway that's behavioral
investment theory and what that does is
it takes you from basic the cambrian
explosion and crabs all the way up
although everything's changing but you
can go all the way up to primates
following this line and that's the
dimension of mindedness we can talk
about the stack but behavioral
investment theory tracks that stacking
the evolution of mind brain behavior in
the animal kingdom it's called mind as a
capital m as really an introduction of a
particular term and that's the first
joint point and so what the descriptive
metaphysics and ontology is oh here's
the set of mental behavior the meta
theory's behavioral investment theory
okay and that goes then from organisms
into animal mental behavior and it's a
joint point that people fail to see too
often so i'll stop there and say this is
then the ground of mental behavior is
we've set up and then we'll go all the
way up through the animal kingdom and
then we'll jump into the human primate
to become a person and what that does
and then finally we actually got to come
back and actually talk about the psyche
but that's a whole nother the whole

another step

so so just to clarify here like you're

you're presenting two things as central

and kind of required for mind with the

capital m

and those are mobility and a central

nervous system is that correct

right it's really it's a complex active

body um

with a central nervous system that ties

up a motor system in a sensory system

together there's interesting an analysis

of evolution about how you get a sensory

system and a motor system how do they

actually evolve when they come together

so it's when you have an exemplar of a

minded animal is a complex centralized

nervous system sent active body that

produces a functional effect on the

animal environment relation

well so how do you rectify

like things like people who are

paralyzed are they then mindless because

they're not mobile

oh of course not so then we need we'll

dialogue exactly so so now we're gonna

then dialogue about the different

domains of mind okay so ultimately we'll

take somebody like you know um stephen

hawking okay so seaman hawking had an

entire shell of existence right i put

through the mediation of the sensory

motor control system on the behavioral

exam to lou gehrig's disease unplugs

that

you know what a nightmare right and then

you get what he then experiences what's

called mind he still has in my language

we can get to it mind one
is inside the organism the
neurocomputational layering and the
modeling inside the organism is called
mind one and that's the neural
computational structure okay if you mind
one a is the inside mind one b is the
mediated by the sensory motor looping
okay
so
definitely you then the entire
definition in the large scale of what
the exemplar is is the sensory motor
loop but obviously then what happens is
what's called mind1a is the
neurocognitive infrastructure within the
nervous system
okay versus mind1b which is the actual
observable from the exterior mediated by
the muscle movements so what happens is
with people that would get this is
mind1b gets x
okay but mind one a the architecture's
still there now if we now jump up to
humans they have what's called mind two
like other primates and i will we can
argue how far it goes mind two is called
subjective conscious experience of being
this first for the hard problem of
consciousness the first person
experience of being mind one gives rise
to then we can dialogue about this mind
two mind two is available from the
inside subject of conscious experience
and then mind three is the justifying
mind which other animals don't have but
humans do okay
and then we created informational

interface with stephen hawking so that he kept after his entire system got shut down instead of locking him in we were able to create a verbal interface so we can maintain contact with his brilliant mind and he could tell us about actually i don't think the information is lost in black holes and i can actually even do the analysis of it okay but if we didn't have that interface he would have been lost but we would have known that yes he would have had mind one a the inside inferential architecture mine two the first person experience of being and mine three a which is the internal narrator but he would have lost mine 3b which is what i'm doing right now talking to you publicly but thankfully we maintained his information interface and uh maintain contact with his great mind thank you greg so um i just wanted to kind of recap two points and just maybe show a little point of contact with what you mentioned in some of the big themes in active inference so first you mentioned how there was these abstractions and mean field approximations in the clinical field and so that didn't that was like speaking to a disconnect between theory and application and practice and making impact on the systems that people care about and then also to the more deeper physical question about mean field approximations and whether you can

study like the trajectory of one
particle versus just take the time
average the space average we're not
going to go into the whole ergodicity
non-equilibrium steady state definition
but those are key terms for active
inference so they come into play with
that difference like i i'm i am i don't
have mastery around the on the active
inference meaning of a lot of those
terms so i need to get i need to be
brought up to speed on some of that but
it's exactly that if someone says well
you know the machine
in the mean field it works but what if
one in 100 fail catastrophically then is
that an approach that works for example
so we've seen that come up in some
machine learning context so that was
kind of a cool um showing how the idea
of the part particular trajectory
versus the mean field and how that plays
out in the way that our knowledge and
application systems health care systems
are constructed which is a niche
modification through us
and then um
so then the second one was like that
combination of appeal to general
frameworks universals
and the historical trajectory of the
whole
horizon of life on earth and of course
our developmental trajectory as
psychological individuals so then it's
like well that looks goal-oriented a lot
i thought goal orientation had to be
purged well i mean the first easy answer

is just that was a period of
disinformation and that
that stance is simply incorrect so
that's one way to simplify the debate
but then
the other maybe longer answer you said
it's energy information fields not
billiards
so we don't appeal to anything when we
see diffusion and ink diffusing now
that's kind of slam dunk but are there
diffusion processes for
metabolism
strategy narrative communication so why
don't we need anything more to just
allow those flows to also happen and so
in active we talk a lot about
those types of models as well
so it's totally a lot of cool areas nice
yup first section yeah no okay so and
the tree of knowledge is both continuous
so it allows for full deflation on the
one hand
or reductive continuity uh into energy
information on the one hand and then if
you look at it it's actually
discontinuous dramatically so in its
funnels
so
matter like mind and culture and so
there's a profound discontinuity as well
and it is the process by if we really
want to get at this it's the process by
understanding this very complicated
dynamic of continuity and discontinuity
and the the unified theory affords a
particular way to understand the
discontinuity of matter out of energy

life out of matter

mind out of life and culture uh out of
mind um and so yeah but but the the
reason when i heard carl you know
especially free energy stuff

although i still am trying to wrap my
head around all the terminology and i
haven't devoted myself to really
uncovering it okay but the the tree of
knowledge basically is a depiction of
negative curvature okay uh and then
the question is how does a system pull
uh

how to via

tracking forms it actually functions to
track forms and make predictions and
it's got to create anticipatory
structures that both
navigate the dialectic of stability and
plasticity to pull free energy into the
structure and i use that term more in an
eric laison way he's a cosmic
evolutionary biologist i mean a
cosmologist

and talks about the evolution of mega
entropy through a free energy principle
um and because i learned that first and
then i was i've been entangled some with
some of the confusion on at least
classic uh call fristens free energy but
anyway it's it's very close i mean i
just i'm sure it's just a technical uh
set of terms i need to download to be
clear about

well

we and many others including anyone who
listens to this are part of the sense
making and we're working on a

participatory ontology there's a lot more to say but let me just give one free energy answer and then maybe let's dive back to a presentation of yours

yep so

um to return to this idea of energy and information fields being primary not billiards

so in a world where matter energy is heterogeneously distributed some is in stars some is in interstellar space or whatever

systems that resist dissipation what we use the blanket terminology to discuss markov blanket systems that reduce dissipation

and have a non-death equilibrium stationary state

those systems are at least acting as if they're trying

which in

complex symbolic ecosystems

means anticipatory predictive behavior

anticipatory predictive signaling and therefore it's helpful to have a action loop model perception cognition action niche impact

it's helpful to have that type of model which is

in the pre-active inference work of karl Frist and others also central the dynamical equations and stochastic um simulations and things like that so there's a lot to say but that we all hope to be resolving that uncertainty in

the country
great i'll pop into where i'm coming
from and this real fast i'll share
screen go to different awesome there's a
um i think i need to go
here hold on one second uh because i
wanted to pull this up
all right
and
did i
maybe i closed it oh well um
yeah
i'll take a second here one second let
me just look at this i thought for sure
i headed up
reset your priors
yep
this is
more complicated than normal let me see
if i got it right here okay
if i go here
and then
one other short thought while you're oh
unless you have okay go for it
i'm gonna just close that down one
second i'm sorry oh yeah take your time
just one other short thought blue do you
want to say anything first
okay um tim bergens questions
so
part of a pluralistic approach towards
understanding includes mechanistic and
historical answers and so the way that
you situated this in the sort of
ecological critique and the
socio-ecological critique the relativist
and not just like nihilist like
relativist things are relational its

interactions all the way down rather
than particles all the way down so
that and the dialectical tradition as
well so many interesting threads so
appreciate how you concisely
discuss that

great

um yeah and i'm happy to pick that up
sorry for my clumsiness here but this is
so here we go in terms of the
basic structure uh that you will
recognize um so here's the i came at
this for what's called perceptual
control theory i don't know if you know
not too many it's underdeveloped
relative to um at least in my fields
clinical psych really doesn't pay any
attention to it well william wt powers
develop perceptual control theory it
comes off the old cybernetic tradition
okay

and all a lot of this will feel uh
similar but anyway behavioral investment
theory then evolves in from a perceptual
control theory model which i see is very
very much in line and essentially what
you're doing here this is the basic
schematic i don't get in i'm not well
versed in mathematics my son's a math
major he teases me all the time i can't
even god damn you have differential
equations which is to my shame but
whatever it is part of my reality um the
bottom line is is here's the basic
schematic okay so you get saline control
variables you get an input transduction
function and you're going to create a
model perception that's going to create

a valence valuing structure that's an energizing action in return so this basic loop is the loop that i've been operating off of and basically it's an active inference uh control loop uh in you know not without getting under the hood of the neural computational specifics about how things are set in terms of its sensitivity um but you can follow this uh at a more sort of psychologically higher level of organization um so that's what i just wanted to show is that it's the the ground of behavioral investment theory is this perception action calibrating loop structure that functions and an example in perceptual control theory is you create an image of an ideal egg and then you run through a bunch of sub goals in relationship to reducing discrepancy towards that you having emotional reactions in relationship to whether or not you're achieving the efficiency of those goals in expected time frames as long as you're not getting surprised and in the constraints and moving in the right direction you're getting jolted with positive affect and continuing the investment if all of a sudden you know you break the egg the wrong way you overcook it there's no eggs in the fridge they can interrupt the structure you get negative activation of affect and then that reboots a possible new behavioral

investment path um so anyway just in terms of sort of the basic architecture of uh that structure that's actually a perfect short section so thanks for making that clear let's get the active loop up so mention uh two points of contact with active and the loop that you described and then blue um so the first one is that in active the preferences are over sensory outcomes so that's very much and i don't know which um if we had a unified knowledge management system theory of knowledge then we could search through which discussions we discussed it but perceptual control theory did come up because preferences are over sensory outcomes so they're not over hidden states of the world which are inferred using latent bayesian models but preferences are over sensory outcomes so that relates very well to the embodied approach and then another just key point of contact which um in the bayesian space it might not feel like this is a major thing but relative to the development of ideas it is and that's related to the experience is perception and the experience of perception is generative so like a neil seth's being you new book summarizing it as well as many other approaches like we're experiencing the generative half

not the sensory incoming half which is
the total

signal processing

info out there model of the world

yep

name in yeah that's that's all part of

so i have uh

um in terms of my own

i've

now comple i don't know if you guys know

john vervicky's work i don't know if his

work has come up at all uh so me and

john have completely unsynced up we

found each other 18 months ago or

whatever and he built um you know

cognitive science and philosophy he

built what's called recursive relevance

realization as a core

structure that builds right into

predictive processing you know and then

the uh what level that like you then

that becomes really refined in terms of

the difference and i certainly get john

here if he's got time he's really busy

but the bottom line is that that john's

recursive relevance realization model

meta theory of cognition and his

taxonomy of knowing in the 40s got four

peers of knowing are now completely

synced up with the unified you talk okay

we did three different

video series one on consciousness one on

the self and one on human transformation

the last with uh zach stein who's a

developmental metapsychologist

educational metapsychologist on expert

transformation so anyway these systems

now are affording all sorts of

productive interplay um in relation so
they now and it's a hand in glove
relation i mean he comes out of
cognitive science and activism and
philosophy i'm coming out of
psychotherapy and
clinical psychology into the science of
psychology and really from a behavioral
and evolutionary perspective as i got
dropped in
neo behaviorals and really it's where my
training is
roots

so anyway then you get so you get an
activism there you know and his stuff
connects so it's a real there's a real
matrix of connection that's actually
operating

blue

nice that's like a perfect flow into um
the question that i had related to both
active inference and in activism
so one of the like fundamental things
that active inference does it kind of
um unifies this inactive embodied
approach as well as representationalist
approaches like where

you know there's mental mental
representation centered around mental
representation um through this like
action perception

loop which is essentially behavior i
mean in some way

and so i just was wondering what has
active inference been fitted to like
this unification of mentalist and
structuralist and behaviorist approaches
or

do you think that that would be an appropriate way to leverage active inference or how else could it maybe be used in your work

well yeah no that's exactly um so i

i totally see that i totally see it as you know the representational and an activism in fact john and i i'm fine with representation john likes modeling as a better term rather than than representation a little uh

he wonders a little bit about some of the history of that term you know it's got it's one of the classic terms that means all these different things but at a generic level i'm totally fine with term representation uh and indeed i'm sort of a neo behavior is coming off an old cognitive simplistic cybernetic model which is basically all representation you've got a mentalist model of representation and then you're mapping the world cognitive maps from you know uh tolman all the way back to that so for me that bridging is very clear and necessary and brilliant and beautiful that an act of inference does um and it's just one of the network possibilities of different kinds of schools of thought

okay

that are afforded and the unified theory basically what i focus on essentially is the class of cognitive approaches weakly defined and then really that the thorn in my side was skinner i was like hello he gets skinner over here okay and at first i just axed him i was like he and

you know he totally negates the agent and information processing from the within in a totally unacceptable way and therefore his philosophy of ryan is wrong after all trump's killed him anyway um so that was my justification then i actually learned what skinner was trying to do i agree with everything i just said by the way but it was also the case that's that skinner was also on to an angle um that red that ultimately sits in psychology as the fundamental mental behavioral kind of schism and ultimately the articulation of mental behavior as the ontology of mental behavior is the way in which you conceptually resolve the underlying disputes in that structure all right so my basic point is yeah 100 we can actually weave all of these schools together now and afford a link a sinking a synergistic linking with coherence that was never afforded before and obviously active inference doing that with representationalism and activism is great it's a piece of the puzzle we can now sync that now that i'm synced in with john this thing all loops back around i'm telling you this thing's about ready to blow we have too many people in uh the active inference field that are also modelers so representationalism is a useful term as opposed to to modeling because it's uh you know it's modeling is like what i do to

of representation or i don't have a
problem

i live in my representations and my
models and they hang out fine together
a lot of

a lot of things i wrote down just wanted
to give one point of difference between
the perceptual control theory and active
inference and maybe some of these other
approaches because i understand that
those are commonly used or at least
known so if someone came to the
you know department of xyz and said you
all know perceptual control theory
perhaps

so

um missing in that loop

was the whole

machine learning type work of policy
selection it just went from basically
like this to inference and then emotion
but it's not just about

how you felt when you looked in the
fridge it's about the multi-joint mostly
sub-personal sub-experiential components
and that's what actually provides the
kind of broad shoulder that connects
cognition across life and maybe even
non-life

is um gives a continuity

to grounded in reduction of uncertainty
conditioned on policy which is what
active formalisms are about
and there's a lot of awesome diagrams
and pros

and theory that's built up in the
qualitative fields and now we have new
tools and new approaches

love it

um

there's some questions in the chat or

you can perhaps present any more slides

well i'll at least just tell the rest of

the story in terms of what uh

what we're so now we've gotten to minded

animals okay complex active bodies

running around after the cambrian

explosion um and they're engaged in the

sensory motor looping battling with each

other for territory control etc okay

then we can i'll be happy to stack that

i basically say that we started off in

the planaria where these reactive

animals okay

and then you get into real learning

animals um and and i believe that the in

the cambrian which are these dynamic

feedback structures that are a lot more

sophisticated and they engage in operant

shaping a good

uh book which is i think right here

uh that i recommend uh the evolution of

the sensitive soul um by uh ginsburg and

jablanca it basically explores the

evolution of that nervous system and

what they call basically a flexible

learning operant system that comes out

of the cambrian explosion it's on

whether or not consciousness gets a grip

in that term meaning when i say

consciousness i mean mind two that's the

phenomenal

that's the ex um nagel's experience of

what it is be like um and basically from

my vantage point you get pleasure

sensory pleasure pain

you know jolts as the first kind of what
john and i call valence qualia okay
probably are the first sets of mine too
and and i'm an early mind to her which
says yeah maybe even down to insects
okay um but that's a huge debate uh we
then climb up the structure of the brain
and then get into a much more extended
inner life
and i certainly think by the time we
have crows and other mammals and things
like that you get a rich mind to inter
perceptual field um
and then they attract that into primates
and you get you know you can defer you
get real relational dynamics this was a
question that came about what about
relation and would talk about that um
but you get relational dynamics with
parenting first and then you get a
parent in a mammal system you get a
parenting attachment system where you're
doing self-other modeling and our care
around another in a long-term relation
and the attachment system comes online
you then get social animals you get
competition and cooperation and you get
this really rich social primate field um
and then i do tomasello's work on the
emergence of a real inter implicit
inter-subjective we space field where
there's actually this capacity for
massive flexible modeling of the other
and mind reading shared attention
intention space this is about 500 000
million years ago when we're you know
pre-fully verbal um you know prominence
running around coordinating to kill

mammoths and like that okay um and so that it's and it's qualitative we're a unique great ape at that time uh because of our implicit group shared attention all of a sudden we can actually like work together to create a coordinated shared community felt sense of what we're doing together this is pre-linguistic and then but what it does is it creates a shared mind space for symbolic tagging okay that's what happens first and music is probably happening etc although that's a good question about when and so now the unified theory picks this up uh at this stage um so there's all the other great work that other theorists are doing and then it says well you get symbolic tagging okay antelope there and you're coordinating the group and that's all very intense requires a lot of definitely high cognitive stuff that's probably pressing the evolution of the brain in a particular way the argument basically is 200 000 years ago probably and we'll see if this gets pushed back but certainly by 50 000 years ago and this is the window here it's a fundamental shift in symbolic communication from a broken symbolic system into a symbolic syntactical system okay they're basically a chomsky and language acquisition device at least in the weak version of that all right that then affords the capacity to place nouns verbs and subjects and objects in a

propositional statement

okay

this is a game changer the proposition

according to unified theory why because

actually a proposition then is a claim

it's a propositional claim about a truth

statement

on the one hand so the antelope are over

there

okay

the antelope are over there then

organizes a particular truth claim that

opens up the possibility for counter

factual negative space

in a radically different way

no they're not

okay those are gazelle not antelope

or and what it does is it opens up the

possibility no we should go hunt rabbits

okay

so propositional claims and the counter

factual structures that they then once

they take a positive affordance and then

the ease with which you can then jump

negative and how do we jump negative off

of a proposition

questions why what who where okay which

are by the way look at one word easy

dynamics

okay

why

all right unbelievably tough to then

answer but it's an unbelievably

interesting on a proposition to add

questions okay so the argument is this

propositions quickly give rise to

questions which then generates what's

called a question answer dynamic

which is the problem of justification

okay justification is creating networks
of legitimizing claims that coordinate
the group

all right

and now this gives rise to then the
evolution of justification systems the
justification systems are the culture
person plane of existence that evolves
out of the primate layer

and they're networked together through
systems of symbolic justification okay
and it makes a very clear prediction
that says okay if that's the case then
there's certain kinds of problems that
come with propositions they're analytic
problems like is it true or false

alternative like what should we invest
in as a group and personal like what
should i share with the group relative
to what i just keep to myself

okay

because the interesting thing about
propositional knowledge is that it
affords a clear window directly into the
mental operations in the mind too in a
way that action doesn't it's like well
tell me what the hell you're thinking

okay

well that's a big goddamn problem why
because you don't want everybody to know
exactly what you're thinking

all right so this is what's called the
el evolution in the unified theory this
is the first key insight happens in 1996
prior to the evolution of the whole tree
of knowledge it was oh my god the human
ego

is the mental organ of justification
okay the secondary narrator on top of
the felt experiential self is tasked
with generating what's justifiable in
terms of what you feel how do you give
reasons to maintain responsibility on
the field and then how do you navigate
your public persona relative to your
private self and filter the dynamics and
relationship
because we do that you do that anytime
you put a little lock on your diary
you're filtering what people can gain
access to okay we're constantly
navigating what it is that people have
access to relative to their justified in
place and believe me now we're a
clinician all the time okay you come
with me the personas out here and their
private cells in here
why because they can't justify what the
hell they actually think and feel to the
world because they'll sacrifice social
influence okay so the dynamic of
justification goes through the skin and
it creates the personal analytic and
social problems which then actually
gives rise to the structure of the ego
which is hey you got to regulate your
experiential self which by the way is
freud's insight your it animal it and
your ego and then your public self and
then internalized introduction called
your superego but then rogers and other
people you know your public face out
here irving goffman from a micro
sociological analysis there are millions
of different places that talk about the

persona relative to the ego relative to
self

and the justification problem then puts
that very in clear evolutionary terms
and what it does is by putting that
puzzle of peace in you're like oh that's
how a synchronized primate which was
some gays and symbolic
took off because the question answer
dynamic exploded this new problem of
justification and that basically hijacks
the language acquisition device into or
associates it with a self-conscious ego
at justifier that then has to navigate
the relationship to the public and the
private

okay and then groups get together and
they build systems of justification
which are what the belief values the
coordinate of population it's like what
do we believe in what do we value in how
do we legitimize whether this guy's
doing the right thing what is is an
ought and how do we coordinate well the
propositional networks of question
answer dynamics build systems of
justification

and that's actually what we live in you
know what we're doing right now is i'm
in sharing with my system of
justification

okay so this thing takes off and then
creates whoa this whole culture person
space up here and evolves it evolves it
right out of the primate
and then that's what humans are they're
both primates and persons
okay and the person space is this reason

giving responsibility so so now what
that means is okay is that there's a
fundamental division remember there's a
division between living animals i mean
living organisms and minded animals
that's a joint point now we have a joint
point between the relationship between
the primate and
the person
which of course we all know there's no
other goddamn primates paying attention
to this talk right now my dog's
wandering around and he's heard this
many different times he still doesn't
get it
okay
so persons are able to get this you have
to live at the level of justification
all right why is that important to get
back to the issue of psychology what is
psychology okay
many people wonder what do you mean by
mental process
and one of the first things i realized
with the subjective pro are we talking
about animals in general
which would be virtually all active
inference stuff
or are we actually talking about persons
with reflective capacities in a social
context that justifies and reasons about
what they do
uh
some psychologists just think in terms
of persons okay other psychologists the
behaviors go all the way down hell the
watson went all the way down to the
goddamn amoebas without a nervous system

okay so the the line of what you refer
to in mental process is very different
and people have different reference
descartes gives us this reason
justifying mind which i now call mind
three so mind three and the culture
person plane of existence is what this
conversation's about mind two is the
felt experience of being and mind one is
the neurocognitive mind one a inside the
nervous system mine one b moving around
on the outside

i got my things just died here i gotta
switch over hold on

ooh

all once he

connects back to the auditory input

then i will continue

then blue

do you have anything first

yeah i have some questions so uh let's

just make sure he's back and then then

go for it all right i'm now back sorry

about that continue perfect perfect um

so i i don't remember where i pulled a

slide away from but but it's kind of

relevant it's one of your slides but

it's relevant to the talk about the

distinction between

self and other and these kind of

complicated dynamics that are layered

into

human human interaction

and you know in this slide that i found

it talks about the language of science

as quantitative third person descriptive

theoretical and public and also then the

language of the subjective self which is

described as qualitative first-person
private um specific
and value-laden and so i just was
wondering if you have thought about
second persons at all maybe you're
familiar with the work of jay garfield
so
just just the relationship between like
like you don't really and jay garfield
like there's the second person
as soon as you realize that there's a
you
then
then you realize that there's a me and
so i think that that's quite fascinating
and i just explained it for people that
maybe haven't heard about it but but um
is there a second person role in your
framework
totally so the slide i think you're
referring to uh hold on i'll get one of
these
thanks
[Music]
let me reach into my exocortex
yeah all of it's all cognitively
offloaded into the niche
um i think the slide you're referring to
has this thing on it it does the iquad
coin and then it has the tree of
knowledge on the other side okay
so all that that slide is saying is is
that science is a particular kind of
language system the thing that the
primary thing i want to emphasize just
in that slide is that science gives rise
to a particular kind of language system
and then i listed all of its components

and i would basically argue that the language of science and what do i mean by that it's like how do we build public knowledge systems that describe the generalizable world what do we mean by sort of really inter-subjective objectivity that then affords a scientific epistemology that then creates things like the periodic table of the elements okay um and it has a particular set of qualities the main thing that i want to emphasize in that slide is if you trace the evolution of science it basically gets more and more it evolves so that it eliminates the unique ideographic particular historical contingent subject at least the emphasis of that and over generalizes or emphasizes the generalizable nomenothetic behavioral law vantage point so and then you can create a dialectic and indeed the unified theory says hey there's a tree of knowledge that's your language of science and then the coin is actually the language of the subject okay so the coin then i got a picture of me on it and it's basically like oh this is my this represents it's actually in the shape of an h and you actually turn it to an i there's an i um and it basically stands for what's called the human identity function okay the human identity function refers to the unique particular subjective

ideographic

epistemological portal that is each one
of us from our particular perspective
that's what it represents okay which is
not a scientific perspective

okay

uh so so so it captures science just
basically factors that out it's
like that's error from science spanish
point okay it's our lives from our
personal vantage point

okay

so now you that your question then is
well do i engage in i mean what do we do
about the second person that's a whole i
totally

in fact the garden is basically a second
person wisdom collective so a second
person is actually how do we create a we
space okay a fundamental shared sense of
being in a collective and what is a
fundamentally about is actually wisdom
is about how do we actually get
collective intelligence together around
an ethic around particular values that
we can coordinate around and feel
fulfilled and so totally it's just not
position there but yeah we definitely
have a second person perspective
very nice

i had a comment on science so um one
distinction that i often think about is
like science as an approach which could
have happened anytime any place then
there's the knowledge slash data the
real specific data sets and then there's
the real specific career and
infrastructure so that just helps me

separate the specific institutions that
we know and love or don't from the data
sets that are good or not from
our approach which is timeless
but then what's interesting about the
slide it says the um is that the
subjective self is the one that's
specific in particular so i understand
the sort of
first and second person side but
actually science as far as it being
generalized and lawful
the counter side is science is the
framework by which we say well we don't
prove anything we don't know that we're
on the back of this animal actually we
have to go look so science does
particularize through measurement and
observation
and the laws of science are probably a
little bit overblown and that's the
whole joke is that people outside of
scientific training don't always have a
sense of what scientists do it's not
alchemy usually
and then that is a disconnect that is
also historical so it's still ground
that's why we need this historical
approach because we have to understand
how does
bacon's early writings
play into
some lay fallacies about what science is
and then won't people just think
differently if they change that belief
beautiful and and now i'll point out
there that yeah so
another key issue here is the meaning of

the word empirical

okay um so the word empirical means
observation systematic observation okay
but early on the people didn't
differentiate between oh i'm i
empirically am observing this purple can
okay

but when science came along and you got
the primary versus secondary quality
dimension that emerged with people like
hume and the whole structure you had the
i experienced this is purple okay
which then they were like well god
that's a goddamn secondary quality which
become qualia okay
and then it's actually what we can
really observe

is the we can measure
and create an inter-subjective objective
intersection with
and that becomes third person empiricism
so the scientist is you have to get
trained in procedures and it becomes a
scientific observation if everybody
trained in the procedure can see and
interpret the data that you gather
so science is not dependent upon a
unique individual idiographic
empirical perspective it's dependent
upon a third person quantitative
generalizable empirical perspective
which is that equal to count as data you
actually have to have transposed
observers that are trained in
interpreting and say oh yeah we all
publicly agree that the way to interpret
you guys look at these goddamn things
coming off with a large hydrogen

collider well we can get together i'll
tell you what that is that's a muon
that's wobbling in a weird way i'll look
at that i don't know what goddamn muon
is but actually all the experts together
and they'll take the observation and say
yep that's a muon and that's wobbling
okay and that becomes an observation so
it's a third-person data-driven
observation that's available to those
that have expertise in the procedures of
interpretation
great one
way that that plays out i think um
it makes me think about how in
frequentism it's sort of like
trial you know accept or reject kind of
and misconceptions beyond that
but with bayesian frameworks even though
the rev base was back in the day
the bayesian framework as it's now
understood in the computational era
helps us deal with probabilistic
variables
and variables we have uncertainty on
and then also allows us which we use a
ton in active is to have like an
underlying process that isn't observed
and then an emitted process that or an
observable that is and having that as
the fundamental that's signal processing
general
partially observable bayesian models so
that's a very established computational
technique and so we can just take it as
our
unit that it's not just the measurements
out there in the world nor is it an

unmeasurable object you say we're always
going to measure the dyad
and that is the unit of scientific
modeling because that's the tools we
have
in unified theory it's observer behavior
relation
cool um okay would you like i mean
there's a lot there's a lot to go there
are we gonna go somewhere else but
anyway there's a foundational it
basically takes science as observer
behavior relation and then it it
actually drops it into quantum mechanics
and general relativity in a particular
kind of way that aligns matrix mechanics
with big bang it's weird but it
basically goes oh yeah the fundamental
unit in fact this is why this is an ipod
coin it's i to the fourth which creates
a unit circle and actually is a
frequency of observer behavior relation
is entangled in the coin so anyway just
as i heard you're saying yep nope that's
exactly what it is and really what
science fundamentally is is observer
behavioral frequency relations that's
what lisa this argument is so i'd love
to hear that
okay uh short thought i hope it's
already an nft but second thought is um
what are the next steps even though
we're not at the end of our conversation
but what are the next steps
for an individual or for the field like
if someone's excited about this
potential
and maybe they're um haven't had a

university education or they have like
what does an individual do to onboard to
know more
to apply or what is the broader project
that you're working on
all right well that's uh yeah so
uh let me see if i can uh
so i gotta ship my frame a little bit
here so let's see what we're talking
about um
so there's the there's a theoretical
angle on this okay so the theor there's
a theoretical angle which to me is that
we're about ready to drop into a new
theoretical enlightenment
okay it's really exciting i really think
that there is an opportunity to address
the enlightenment gap
and find a way to place the sciences in
proper relation to our knowledge and do
so with consilience that's super
exciting from a theory perspective okay
but it turns out and and so if that was
going on that's beautiful and wonderful
okay but i have to pause because that's
not the only thing that's actually
happening here at least from my vantage
point
um
and that's the world is happening
okay
what do i mean by that i mean like the
world is like in a very interesting spot
right now
okay and is happening at a very
very um weird level okay why is this
relevant for me well i'll just say okay
so

you know i've been in the academy my whole life all right i've been i've i got i was at the university of pennsylvania worked for beck learned all that you know published its special issues blah blah trying to get the arc the institutional infrastructure to upgrade itself uh at least but i couldn't there was no i didn't i came at it from a two different position um and ultimately although i was inside the institution i couldn't get any leverage um and ultimately over the last four years things have happened so that i have jettisoned myself outside the i'm still full professor and i do my job okay but actually a wave has happened inside the institution that i feel qualitatively more alienated from over the last four years and that's for personal reasons okay and retrospective reasons in relationship to um my difficulty trying to get traction and global reasons and zeitgeist reasons so that now i hang out with people in that are futurists okay so my my i used to spend you know in traditional blah blah and i still do but i actually spend all my time talking with people that are in the midst of a zeitgeist like for example i'm involved in like metamodernism okay metamodernism is an emerging sensibility okay that is basically the simplest way

to think about it is the thing that
comes after post-modernism
that looks back and resolves the
modernist versus postmodern
modern thesis post-modern antithesis
meta-modern synthesis
okay and there's a emerging sensibility
in europe about this which emerges over
the last decade okay
and weird happens with that so like
the an ethos is sort of like it's a it's
seeking a post-post-modern grand meta
narrative so post-modernism says there
is no grandma narrative they're always
dangerous and this one comes after and
says i wrote in my book in 2011. and i
was like hey the unified there is a post
post-modern grand better narrative
and then i turned this into a garden
and i don't know why the hell i did that
you know that's weird okay
but the issue is is that when i then saw
the artistic sensibility of
metamodernism it's called sincere irony
okay sincere in the sense it's authentic
and true irony in a sense it's a
critique and twisted well that's what
that is this is unbelievably sincerity
and it's a cartoon and it's twisted and
it's both science and religion and it's
a joke and it's serious and it's all
that i had no idea that the meta-modern
sensibility had a sincere ironic
flavor i was called to do this okay oh
i'm connecting to people now the
experience that i have is like the the i
don't know if you saw the close
encounters are the third kind

all right close encounters occurred kind of like 10 000 people in the world sort of wake up and they get this image of devil's mountain all right and he gets his callings crazy calling like oh my god oh my god and then everybody migrates to it and then blow the whole that's where the aliens show up okay well i'm telling you i feel like there's a calling that's happening for in the process of a particular serious evolution we're in a serious evolutionary flux okay and and the tree of knowledge actually clarifies exactly why we would be remember each one of those life mind culture what does it say each one of those is an information processing and communication network system that yolks stuff together at a higher order huh okay so cells did that and then animals did that and then language did that is it any possible that there could be an information processing network communication system that is emerging at a higher order well you actually do that you look back and you're like oh my god that's a goddamn 20th century we laid down the internet and we built artificial intelligence systems that are intersecting with it okay so from a tree of knowledge perspective we're actually in the midst of the fifth joint point on this century this century

is the fifth joint brain when you're
traveling through the culture person
plane of existence into the potential
digital it's a whole new potential
metaverse of complexification
okay if you see the world through that
particular ends then you're like whoa
now we can actually track the evolution
of complexification across all these
different frequencies levels with at
least intuitive comprehensibility and
you can then see well we're
actually in the midst of a flux point
as we have built all of these different
technologies that transcending the old
infrastructures for institutional
control and regulation sense making and
we're all going insane because this is
opening up all sorts of possibilities
but it's also threatening the
fundamental foundation of how we operate
and then the question is can we upgrade
our wisdom to navigate the evolution of
technology and the planet and do so in a
way that stabilizes
or is there going to be too much goddamn
chaos and flux the system's going to be
thinly stacked everybody gets
defensive and insane because of the new
thing and the goddamn thing plummets
blue buy sell
yeah intense i mean i don't know
i'm lost in the in the artificial
intelligence and cognitive offloading
and niche modification i think i i
wonder about what you feel like the role
is
since you brought up the internet um and

artificial intelligence like what do you
feel like the role is of um niche
modification in the transmission of
culture or do you feel that culture is
something that has to then be
transmitted directly like is that a
person to person a learned behavior or
can we modify the niche
and transmit it in that way
well so
let's to be clear about some definitions
so in the true knowledge it's capital c
culture okay then there's lower c
culture but capital c culture is the
technical term for the systems of
justification
the propositional networks at various
levels that legitimize is an ought in
inside groups or outside of them so if
you're inside a culture you have a
propositional network of legitimizing is
an art that can be coordinated with
yourself at the ego level diatically
or move on up the stack so if you take
act inference lab and you create a
culture on it it will be networked
around a system of justification that
legitimizes what it is that you're doing
in a particular niche and that creates
the act inference culture
capital city propositional network
there's little sea culture which is
basically the behavioral investment
patterns other animals have little seed
culture okay that then afford the
capacity to adjust into niches and then
feed back on them you know animals learn
how to work on washed potatoes no

monkeys will do that all right um and
you can then and of course we're
unbelievably flexible in our behavioral
repertoires all right
so now the fundamental issue is that we
are then uh from a macroscopic level
what are the sys what are the
large-scale systems of justification
that are operating
to coordinate is an awe
and the answer is not much because of
the chaos in the globe the globe has
afforded us to get together but our
knowledge systems are not up to the task
of coordinating we know where to learn
it this is the metaverse i mean what the
hell is this thing okay and so we're
trying to we're groping to build
justification systems that can
effectively hold and cultivate all the
niches that actually then need to grow
into that afford individual variation
and new growth and blah blah and at the
same time
do so in a way that you know well makes
this transition or at least affords the
emergence of this whole new way of
relating the digital okay that can
collapse space and time and have us talk
to each other at the exact same time
which is totally weird right um
so
i guess my basic point is there's
multi-level
um
operations all the time okay uh that are
down more trans-determinative meaning
there's everything is sort of happening

across the synaptic diachromatic stack
okay well i'm not a rigid determinist by
any stretch um
but anyway then our i see us as having a
moment of conscious evolution like the
issue fundamentally for us um at a broad
level is to wake up to the situation we
find ourselves in
okay and upgrade our operating system to
be up to the task upgrading our system
of justification
to gain knowledge and socialization so
we can be effectively oriented toward
wisdom that's what the fundamental
garden is basically there is a way um
that we can then and then if we solve
some of the old problems that have
prevented this upgrade from happening so
that we can see the path to the upgrade
then a map of cognitive architectural
map from the sort of the intellectual
stream will happen if that's in good
contact with people that are realizing
that the system is sick right now and
want a community upgrade then the
intellectual solution can feed the
community upgrade they can metabolize it
and get it to normal people living their
lives and we can change the contingency
infrastructure nation state capital
labor relations resource control stuff
and hopefully then evolve so to me the
issue is very flux
very bottom up top down
lots of different things will influence
at the same time
my goal is really there's a meaning in
mental health crisis that's one aspect

of the dynamic we face
that's what the unifi you talk
fundamentally then it's oriented towards
so my life mission is we can do a lot
better in meaning making and mental
health than we currently are
there are ways to understand knowledge
and wisdom that can fundamentally shift
and rotate the way we think about
ourselves so we can in fact i'm
searching for a coherent naturalistic
ontology that can revitalize the human
soul and spirit in the 21st century
that's essentially what my the mission
is and utah affords a particular
sense-making structure to do that and if
everyone's operating in that kind of
sense-making structure from both the
bottom and the top the whole coordinated
system of justification would be
different and the consequences of many
many things would be different
thanks for sharing that
just to connect it to two or a few
active ideas
policy selection which is core
is
based upon the partitioning but always
the joint consideration of epistemic
value knowledge oriented value or
intrinsic motivation and pragmatic value
utility
external motivation so we need to
learn more you know we got to take a few
steps back to go the easier way to the
mountaintop but also we need to clear
the clouds to see where we could go and
communicate how we could go so that's

why planning as inference is a useful
framework because it's not enough to
just have preferences preferences are a
core part of our models we want to be
like warm and secure
at each level of the maslow's hierarchy
but the other side that in activism kind
of slapped back into the world
we have to have a plan
and it has to be pragmatic
so
just beautiful very
similar
um
it's funny because
drawing more from the cybernetics
and
dynamic systems and complex systems
modeling side
the experience
that's the part that's oh that's
unsupported or you'll never be able to
make those claims or that's you know
that's the pl gangplank
versus the kernel which is seen as to be
the math but then the real relational
insight
shows that it's truly
a bi-directional freeway like everyone's
going home
or away because you really can wrap
a lot of the technical parts within a
broader picture
that isn't formal
amen brother
so just a quick question uh related to
both the mental health crisis and to
culture with the capital c

um which

i don't know

is in that kind of definition like

something i'm equating to like a moral system or a value system is that is that

appropriate yeah oh sure yeah it's a

it's a belief value network of

propositions that afford a group to be

able to state explicitly what is an awe

and negotiate from there

so what do you think um

or do you think that there is a

relationship between

like the fact that you know in western

society at least in the states like many

of us are just spiritually bankrupt

and so like do you think that there is a

connection is that a possible

fundamental reason for the mental health

crisis that that we're facing now do you

think that those are linked exactly no

directly 100

yeah no no we we we've we're operating

off of a very sick system

of justification and and um it's

basically my friend zack stein puts it

this way um or at least i'll draw from

him and i'll put it in my own basic

context so what happens with the

modernity which is basically

enlightenment capital labor relations uh

democrat and then the evolution and what

modernity does to the traditional

knowledge systems and the oral

indigenous knowledge systems prior to

that okay is it basically shifts the

structure into a skill development um

self-interested control model okay

basically so now the values of justification that evolve off of the modernity are not spiritual insolent but ultimately reductive physical self-interested control mechanistic kinds of instrumental ways of being okay which is an important way of being no doubt um but it's a doing problem okay which is different than eric from we talk about as a being problem all right let me give you an example which i stumbled into through the unified theory to explain this okay so the fourth branch back there is this thing called the influence matrix all right and what it does is it basically combines attachment theory and the interpersonal circumplex uh to give a model of the process dimensions that we track in our dyadic relational world okay meaning the self-other world and then we wander around with internal working models in relationship to it okay so the the core line in the influence matrix i originally called uh the social influence line that's why it's called the influence matrix and it said that we had high social influence or low social influence okay and influence here was defined in pretty instrumental terms the instrumental terms were the capacity to influence others in accordance with your interests or the reverse the absence of that so there's a resource in conception of influence and it's the capacity and that's becomes the fundament and from a pure

kind of a standard evolutionary
mechanistic modernist kind of view this
capacity to influence the world is where
rubber meets the road okay
but over time that thing evolved and it
became the relational value and social
influence line

okay

whereby what we're actually tracking is
both our capacity to move other people
instrumentally and the extent to which
we're known and valued by important
others

okay and that is our heart our heart is
both hey can we actually move people
around

and be am i genuinely known and valued
by important others for my essence

okay and what this emerges in my
clinical work okay is a i work some with
some people who had no social influence
and no relational value homeless people
who are brutalized by all aspects of
life and they're just miserable

okay

but i then met with a number of people
of course this will come as no surprise
who actually had high social influence
who were completely empty
they manipulated and controlled other
people but they didn't they had they
stacked themselves up with high resumes
but then nobody liked them they could be
bosses and control people and
had money

okay so they had instrumental influence
but and of course then if the instrument
just changes you now have to spend all

your time especially if you're going to
do it through control to gain influence
through power

and now that's a very vulnerable spot
you lose access to control and you're
jettisoned

okay

and then there are other people even who
were actually valued by other people but
they put on a face their persona is what
they felt was value they weren't known
okay and these people also felt very
anxious because they could be discovered
for who they were

the core heart placement of security is
to have social influence and be known in
value for your essence

okay hunter-gatherers get this

we because of our instrumentality in the
way we decide to control then we create
economic labor relations we basically
turn everything into an influence game
and we lose relational value

okay we don't know that actually that's
really what i mean uh you know john
watson basically decided that the reason
kids wanted hugs was because it was
paired with food i mean that's how
insane his model of humanity was and by
the way two of his three kids killed
themselves

that's john watson behaviors okay so our
idea of self-interested mechanistic
control and applying the clockwork way
in which power force mass and
acceleration actually work as if really
that's the underlying causation of our
true essence actually kills our soul

because it turns it into instrumentality
and fails to attend to the genuine
beingness and the value of that we have
to revitalize that and we know it when
we feel it okay but our justification
systems
meaning our academic hard-nosed
tough-minded thing was basically well
when it really comes down to it you're
basically just a bunch of molecules i
mean that was the message okay and we
need to change that message people
digested that message people lost the
clarity about what a human soul and
spirit is from a naturalistic
perspective we can revitalize that get
clear about it point them to it and if
we build systems of justification that
then organize institutions like
education politics national relations et
cetera the whole goddamn change equation
changes at least according to this
perspective
very nice
so i it just brings me to another
question that i had unless daniel you
have something that you wanna about blue
with the last question and then we'll
have a closing thought so thanks a lot
greg this is really a cool dialogue so
blue last question and then we'll each
give a last thought
so um just one final question is
speaking of you know relational
interactions and you know if the
relational nature isn't food and it
isn't like we're not billiard balls
we're existing in you know a field

essentially like what is the relational
language like how can we best
relate to one another
relate to ourselves like how do we speak
that dialogue beautiful
um well so i built this thing called map
of mind to just defer metaphysics of the
mental domains and there are five
domains that are very obvious okay so
one is mind1b that's
okay mind1a that's what my nervous
system is doing okay mine two that's the
i can see but you can't
mine three a is what i talk to myself
okay i was just doing that
mind3b is now me talking out here okay
so and those are all easy any behavioral
no matter how empirical they are you can
now have a map that says oh these are
actually the meanings of referent of
mind and notice there are five of them
and we use the word mind and actually
there are five different reference here
we need to get very clear because
they're different epistemological and
ontological reference points okay
well
here's one that um gets right to your
point okay
i think and i think michael tomasello's
work and i think anybody that really
falls in love
and feels is mine to be
okay
the inter-subjective non-verbal
intersubjective space of tracking one
another and participating in the
attachment relational dance of being

it's seeing the other person through the
eyes tracking where their eyes are and
essentially seeing their heart their
felt experience of being and being part
of that and then being in dance with
that and if anybody's ever fallen in
love with somebody
okay and then you fall into that and
then all of a sudden you are that field
that felt experience of being
you know joins together in a particular
way and i actually think there will we
can develop you know genuine
metaphysical claims about mind to be
space which is an implicit inner
subjective space i think tonicello's
work on shared attention and intention i
think many different things domain in
relationship to that and it basically is
a relation a pre-verbal primate
relational field that we are actually
really attuned to and i think that that
informs us a way to think about how we
sync up with each other relationally i
think we should shine a light on that
because i think we're relationally
deprived and i think that many things
like social media basically break that
completely relative to what it is
designed to track and so you're seeing a
mental health crisis in adolescence they
don't have any mind to be connection
everything is texting and likes
and you know jolting and everything else
and the whole sense of who i am and who
you are i mean you go back to hunter
gatherers you lived in a family forever
and they saw all of this it was real

world

action

i don't know and i think that's not what
the heart needs and some that we got to
get we got to figure out a way uh to
cultivate that or else we're going to be
empty and so yeah let's shine a light on
that mind to be relational world
it's either going to be 2b or not to be
brilliant thank you

touche or whatever really at least love
it

i'll give a last comment or blue do you
want to go first and then to greg for
the final

um i think that this was you know
especially the last remark was just very
beautiful and very poignant um so thanks
greg for for coming on and it's been a
pleasure

okay

see you later but really any time that
first off if anyone wants to get
involved in active lab hopefully even in
the future if you're watching this we're
still going to be there to help scaffold
and um yeah greg if you or any
colleagues or anyone else ever wants to
get involved or do another discussion or
anything else this was really
a great convo

beautiful well thank you so much it's a
really honor to be here i really enjoyed
it um i'm interested in us cultivating
wisdom energy that's that's one way of
summarizing what the project is here and
uh for folks that you asked like what
can we do

ultimately yes look out have some
concern look around at your local garden
and try to cultivate wisdom energy in
your local garden with your heart
yourself and the other people that you
love in the world and we do that enough
when we might be all right
and we have the ontology working group
in the lab
for when that's not all right right well
this and i and as a theorist that's
absolutely essential the ontology the
failure of ontology enlightenment gap is
part of why we're in the mess if
we can get this ontology right we can
get on the right path 100
thank you greg talk to you later see you
blue thank you brilliant take care okay